

3 TYPES OF COMPUTERS

There is a variety of computers of different power, size, function and designed for different purposes. They can be divided into several categories and are often **classified by size, power and portability**. The most common types by size and power are supercomputers, mainframes, personal computers (desktops), portables and mobile devices.

The fastest and most powerful computers are called **supercomputers** which can process one quadrillion of operations per second. A supercomputing centre was founded at the Technical University of Ostrava called IT4Innovations.

introduced in 1984 was the first computer with a graphical user interface. **Servers** are special computers providing various services and often placed in a rack (rack-mounted). They are classified according to their purpose, e.g. web server, file server, database server, mail server, etc. Workstations are high-performance desktop computers for technical and graphical applications, etc.

Portable and mobile computers are smaller than desktops, e.g. laptop, notebook, subnotebook, netbook, tablet, etc. Laptops have the compact design, flat panel display, a touchpad built into the keyboard dock. Ports for

Computer cluster



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Mainframes are large, powerful, expensive computers that are operated by a team of professionals. They are used by many people at the same time (multi-user systems) for processing very large amounts of data and performing multiple tasks. Mainframes are used by large organizations in banking, airlines or government.

Desktops have their own operating system, processing unit, monitor and keyboard. They can also be used as a workstation or a smaller server. A computer case can be oriented horizontally or vertically (called a tower). Macintosh PC produced by Apple Computer Inc. and

connecting peripherals and slots for memory cards are placed in the front, side and back panels. Laptops can be powered from rechargeable batteries.

A **tablet** is a modification of a laptop. It is a full-function light-weight computer which has a touch-sensitive screen and on-screen keyboard. It is operated by mobile OS (Android, iOS, Windows). Portables use wireless technologies (Bluetooth, Wi-Fi, cellular networks).

Smartphone is a mobile multimedia device with a mobile OS. It combines the features of a cell phone with a camera, a media player and an internet device.

It has processor, memory, a touchscreen user interface, high-speed mobile broadband (data transfer capability, transmission over a wide range of frequencies), wireless Internet access, a lot of sensors that measure motion, orientation and various conditions. It uses a battery charger or power bank. Phones transmit and make calls through base stations, transmitting towers or antennas forming cellular or mobile networks. They also enable Internet phoning (Voice over Internet Protocol). A phablet is a combination of a smartphone and a tablet.

Microcontrollers are tiny computers that enable the user to store data and execute simple commands and tasks. There are known as embedded systems, for example Arduino. Very small computers can be embedded (inserted, fixed) or they can be wearable on human body.



VOCABULARY

base station /'beɪs ˌsteɪʃən/	základnová stanice
battery charger /'bæt.ər.i ˈtʃɑːdʒər/	nabíječka
broadband /'brɔːd.bænd/	širokopásmový přenos
camera /'kæm.rə/	fotoaparát
case /keɪs/	počítačová skříň
cell phone /'sel fəʊn/	mobilní telefon
desktop PC /'desk.tɒp ˌpiː 'siː/	stolní počítač
embedded /ɪm'bed.ɪd/	vestavěný
laptop /'læp.tɒp/	notebook
mainframe /'meɪn.freɪm/	mainframe (dříve sálový počítač)
on-screen keyboard /ɒn'skriːn ˈkiː.bɔːd/	virtuální klávesnice
personal computer /ˌpɜːsənl kəm'pjʊːtər/	osobní počítač
portability /ˌpɔːte'blɪtɪ/	přenosnost
portable /'pɔːtə.bəl/	přenosný
power /paʊər/	výkon
powerful /'paʊə.fəl/	výkonný
server /'sɜːvər/	server
size /saɪz/	velikost
smartphone /'smɑːt.fəʊn/	chytrý telefon

supercomputer

/ˈsuːpəˌkəmˌpjʊːtər(r)/

superpočítač

tower

/taʊər/

věž

transmission

/trænz'mɪʃ.ən/

přenos

wearable

/'weə.rə.bəl/

nositelný

workstation

/'wɜːk.steɪ.ʃən/

pracovní stanice

TASKS

TASK 1

Comparing different types of computers

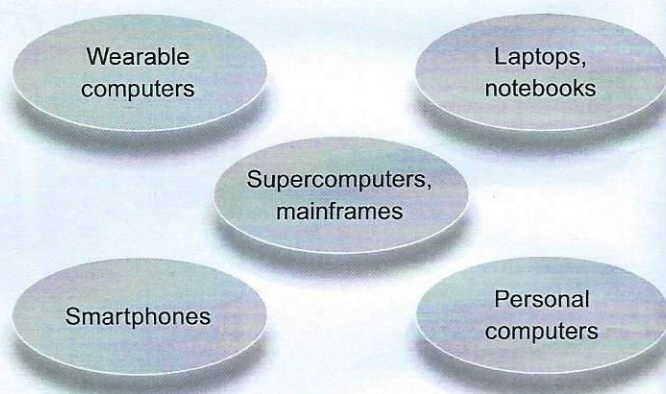
To make a comparison between two or more things, we use comparative adjectives with *-er* or *more/less* + *than* and superlative adjectives with the definite article. Superlatives are formed by adding *-est* or with *the most/least*. Some adjectives have irregular forms.

Example:

Desktop PCs are heavier than laptops.

Supercomputers are the largest and the most expensive computers.

Work in pairs or small groups. Choose one type of computers (see picture), describe it and then compare with other types mentioning advantages and disadvantages.



ACTIVITY

(Web-based)

Find out more information about the National Supercomputing Centre IT4Innovations in Ostrava with the help of this website: <http://www.it4i.cz/?lang=en>

QUESTIONS

- Which type of computer is the most powerful one?
- Which type is suitable for people who travel a lot?
- Find what do "all-in-one" computers and "2-in-1" computers (hybrid laptops, convertible laptops) mean?
- What is a server / a workstation?
- Which type(s) of computer do you use and why?